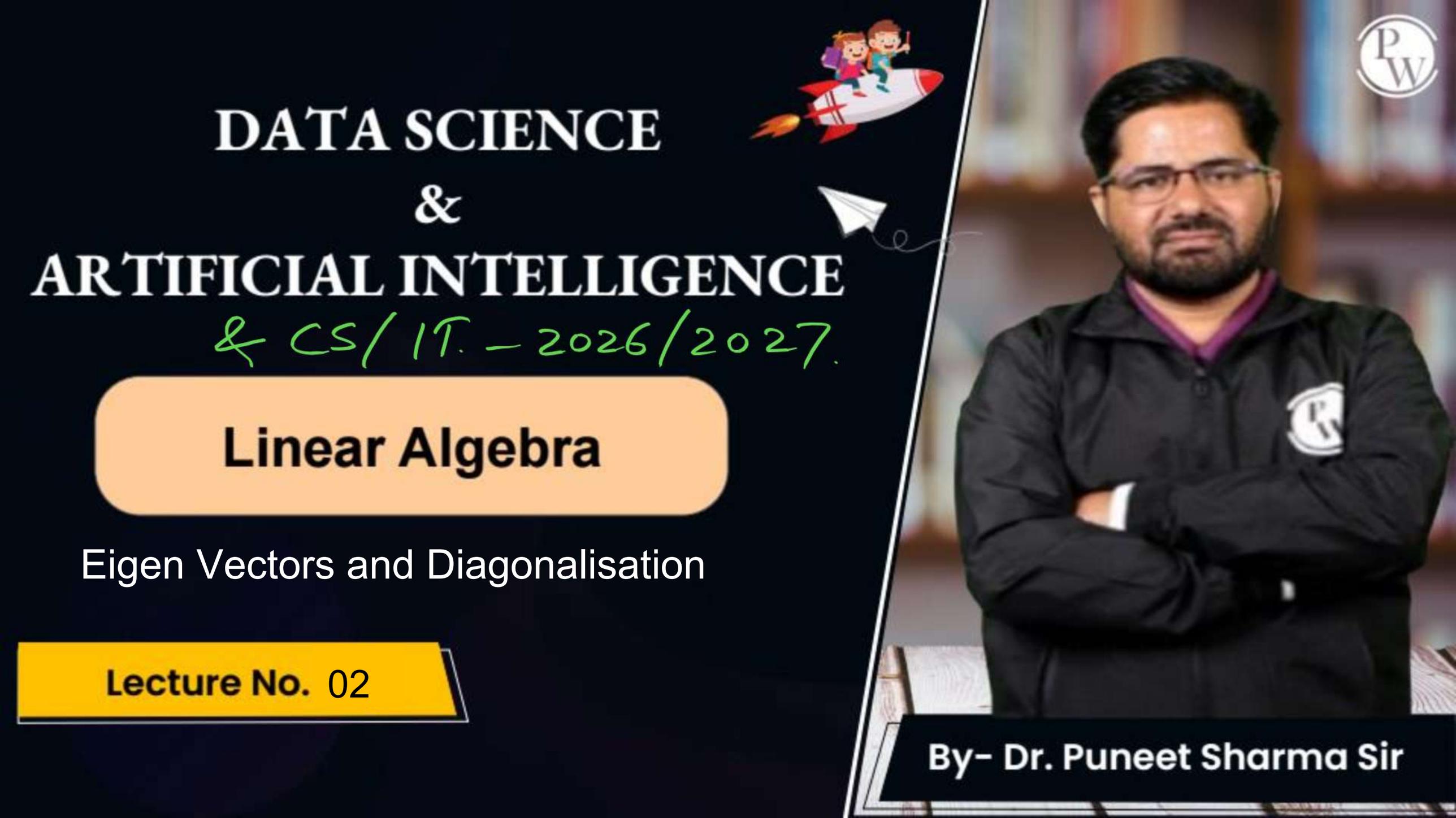
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Eigen Vectors and Diagonalisation

Lecture No. 02



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

EIGEN VECTORS

Topics to be Covered



Topic

1. Remaining Part of Eigen Vectors
2. Concept of Diagonalisation
3. Rotation Matrix

PROPERTIES of EVECTORS →

① we are free to Multiply or divide with any Constant in Case of E.Vectors.

$\because AX = \lambda X \Rightarrow \boxed{(A-\lambda I)X = O} \Rightarrow K(A-\lambda I)X = KO \Rightarrow \boxed{(A-\lambda I)(KX) = O}$

① — Non Zero E Vector — Non Zero Soln ≃ ∞ Soln

② — Non Zero E Vector

i.e whenever X is an E.Vector ⟺ (KX) will also be an E.Vector

② E Vectors for different E Values are LI

③ If E Value Repeats (then Headache starts) then Corresponding E Vectors may be LI or may be LD.

(4) **MODAL MATRIX:** MATRIX formed by E.Vectors is called Modal Matrix.

Let $A_{3\times3}$ & it's EVectors are X_1, X_2, X_3

then Modal Mat $P = [X_1\ X_2\ X_3]$

if $|P| \neq 0 \Rightarrow$ EVectors are LI & if $|P| = 0 \Rightarrow$ EVectors are (LD)

(5) E.Vectors Corresponding to different E.Values of Symmetric Mat are orthogonal

(2015) Consider Symm Mat $A_{3\times3}$ s.t it's E.Values are a & b & Corresponding EVectors are $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ & $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ then for $a \neq b$ the value of $(x_1y_1 + x_2y_2 + x_3y_3) = ?$

$X =$ $Y =$

By above prop: $X \cdot Y = 0 \Rightarrow x_1y_1 + x_2y_2 + x_3y_3 = 0$

⑥ Algebraic Multiplicity (A.M) = Number of times λ Repeats is called AM of λ

Geometric Multiplicity (G.M) = Counting of LI E.Vectors for λ is called GM of λ

Shortcut of finding GM → Let λ is the Repeated E.Value then

$$\text{GM of } \lambda = \text{order} - \rho(A - \lambda I)$$

ie Number of L.I E.Vectors for λ = order $- \rho(A - \lambda I)$

(1) Shortcut Method of finding E.Vectors when options are given

" λ should be unique in all comparisions & their should not be any INVALID Result " Then in the eqn $AX=\lambda X$, X will be E.Vectors.

Eg: which of the following is not an E.Vector for $A=\begin{bmatrix}1&2\\0&2\end{bmatrix}$

(a) $\begin{bmatrix}5\\0\end{bmatrix}$

(b) $\begin{bmatrix}1\\0.5\end{bmatrix}$

(c) ✓ $\begin{bmatrix}-3\\-1\end{bmatrix}$

(d) $\begin{bmatrix}4\\2\end{bmatrix}$

Taking (a): $AX=\lambda X$

$$\begin{bmatrix}1&2\\0&2\end{bmatrix}\begin{bmatrix}5\\0\end{bmatrix}=\lambda\begin{bmatrix}5\\0\end{bmatrix}$$

$$\begin{bmatrix}5\\0\end{bmatrix}=\begin{bmatrix}5\lambda\\0\end{bmatrix}$$

$\lambda=1$, $0=0$ ✓

unique, Valid

Taking (b) $AX=\lambda X$

$$\begin{bmatrix}1&2\\0&2\end{bmatrix}\begin{bmatrix}1\\1/2\end{bmatrix}=\lambda\begin{bmatrix}1\\1/2\end{bmatrix}$$

$$\begin{bmatrix}2\\1\end{bmatrix}=\begin{bmatrix}\lambda\\\lambda/2\end{bmatrix}$$

$\lambda=2$, $\lambda=2$ ✓ unique

(c): $AX=\lambda X$

$$\begin{bmatrix}1&2\\0&2\end{bmatrix}\begin{bmatrix}-3\\-1\end{bmatrix}=\lambda\begin{bmatrix}-3\\-1\end{bmatrix}$$

$$\begin{bmatrix}-5\\-2\end{bmatrix}=\begin{bmatrix}-3\lambda\\-\lambda\end{bmatrix}$$

$\lambda=5/3$, $\lambda=2$ ✗ Not unique

Q. E. Vectors of $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$ is/are ?

MSQ

(a) $\begin{bmatrix}5\\5\\0\end{bmatrix}$

(b) $\begin{bmatrix}-3\\0\\3\end{bmatrix}$

(c) ✗ $\begin{bmatrix}2\\1\\0\end{bmatrix}$

(d) ✗ $\begin{bmatrix}4\\2\\1\end{bmatrix}$

(a): $AX=\lambda X$

$$\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}\begin{bmatrix}5\\5\\0\end{bmatrix}=\lambda\begin{bmatrix}5\\5\\0\end{bmatrix}$$

$$\begin{bmatrix}15\\15\\0\end{bmatrix}=\begin{bmatrix}5\lambda\\5\lambda\\0\end{bmatrix}$$

$\lambda=3, \lambda=3, 0=0$

unique — Valid

(b) $AX=\lambda X$

$$\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}\begin{bmatrix}-3\\0\\3\end{bmatrix}=\lambda\begin{bmatrix}-3\\0\\3\end{bmatrix}$$

$$\begin{bmatrix}-3\\0\\3\end{bmatrix}=\begin{bmatrix}-3\lambda\\0\\3\lambda\end{bmatrix}$$

$\lambda=1, 0=0, \lambda=1$ ✓

Similarly (d) is also not an E. Vector.

(c) $AX=\lambda X$

$$\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}\begin{bmatrix}2\\1\\0\end{bmatrix}=\lambda\begin{bmatrix}2\\1\\0\end{bmatrix}$$

$$\begin{bmatrix}5\\4\\0\end{bmatrix}=\begin{bmatrix}2\lambda\\\lambda\\0\end{bmatrix}$$

$\lambda=\frac{5}{2}, \lambda=4, 0=0$

Not unique — Valid

So (c) is not an E.V.

Q. one of the E vector of $A=\begin{bmatrix}5&0&0&0\\0&5&5&0\\0&0&2&1\\0&0&3&1\end{bmatrix}$ will be?

(a) $\begin{bmatrix}1\\2\\-3\\0\end{bmatrix}$, (b) $\begin{bmatrix}1\\-2\\0\\0\end{bmatrix}$, (c) $\begin{bmatrix}-2\\0\\1\\2\end{bmatrix}$ (d) $\begin{bmatrix}-1\\2\\1\\2\end{bmatrix}$

(ii) Also find Corresponding E. Value

Taking (a) $AX=\lambda X$

$$\begin{bmatrix}5&0&0&0\\0&5&5&0\\0&0&2&1\\0&0&3&1\end{bmatrix}\begin{bmatrix}1\\2\\-3\\0\end{bmatrix}=\lambda\begin{bmatrix}1\\2\\-3\\0\end{bmatrix}$$

$$\begin{bmatrix}5\\-5\\-6\\-9\end{bmatrix}=\begin{bmatrix}\lambda\\2\lambda\\-3\lambda\\0\end{bmatrix}\Rightarrow \underbrace{\lambda=5,\lambda=\frac{-5}{2},\lambda=2}_{\text{Not unique}},\ \underbrace{-9=0}_{\text{Invalid}}$$ is (a) ✗

Taking (b) $AX=\lambda X$

$$\begin{bmatrix}5&0&0&0\\0&5&5&0\\0&0&2&1\\0&0&3&1\end{bmatrix}\begin{bmatrix}1\\-2\\0\\0\end{bmatrix}=\lambda\begin{bmatrix}1\\-2\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}5\\-10\\0\\0\end{bmatrix}=\begin{bmatrix}\lambda\\-2\lambda\\0\\0\end{bmatrix}$$

$\underbrace{\lambda=5,\lambda=5,}\ \underbrace{0=0,0=0}\rightarrow$ (b) ✓

(ii) & Corresponding E Value is $\lambda=5$

Similarly we can show that (c) & (d) are not E vectors

(M.II) Using Conventional APP →

$$A=\begin{bmatrix}5&0&0&0\\0&5&5&0\\0&0&2&1\\0&0&3&1\end{bmatrix}$$

CEqu^n is $|A-\lambda I|=0$

---------- (10 mins)

$\lambda=\lambda_1,\lambda_2,\lambda_3,\lambda_4$

E. Vector for (λ_1) → $AX=\lambda_1 X_1$

$(A-\lambda_1 I)X_1=0$

10 mins

TOTAL WASTE of TIME.

$X_1=\begin{bmatrix}\\\\\\\end{bmatrix}\neq 0$

E vectors for λ_2 → $AX=\lambda_2 X$

$(A-\lambda_2)X=0$

10 mins ---- $X_2=\begin{bmatrix}\\\\\\\end{bmatrix}\neq 0$

(93) (MSQ) $A=\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}$, $\alpha\neq 0$ then Eigenvectors are ?

(a) $(\alpha\ 0\ 0)^T$

(b) $(0\ 0\ 1)^T$ ✗

(c) $(0\ 0\ \alpha)^T$ ✗

(d) $(0\ \alpha\ 0)^T$

(a): $AX=\lambda X$

$$\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}\begin{bmatrix}\alpha\\0\\0\end{bmatrix}=\lambda\begin{bmatrix}\alpha\\0\\0\end{bmatrix}$$

$$\begin{bmatrix}0\\0\\0\end{bmatrix}=\begin{bmatrix}\lambda\alpha\\0\\0\end{bmatrix}$$

$\lambda\alpha=0,\ 0=0,\ 0=0$

$\lambda=0$, Valid ✓

(b) $AX=\lambda X$

$$\begin{bmatrix}0&0&\alpha\\0&0&0\\0&0&0\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}=\lambda\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$\begin{bmatrix}\alpha\\0\\0\end{bmatrix}=\begin{bmatrix}0\\0\\\lambda\end{bmatrix}$$

$\alpha=0$, $0=0$, $\lambda=0$

Invalid, Valid, unique

Similarly (c) is wrong & (d) is correct.

Q. Number of L.I E.vectors for $A = \begin{bmatrix} 3 & 0 \\ -5 & 4 \end{bmatrix}$ will be ? = Two.

∵ A is LTM so $\lambda = 3, 4$

∵ Both the E.values are different, & corresponding E.vectors are (LI)

Q. one pair of (LI) E.vectors of $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$

(a) ✗ $\begin{bmatrix} 5 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \end{bmatrix}$: LD

(b) ✗ $\begin{bmatrix} -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \end{bmatrix}$: LD

(c) ✗ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

(d) ✓ $\begin{bmatrix} 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}$

Taking (c): $AX = \lambda X$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \lambda\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2\lambda \\ \lambda \end{bmatrix}$$

$\lambda = 2, \lambda = 2$ ✓

& $AX = \lambda X$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \lambda\begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 13 \\ 10 \end{bmatrix} = \begin{bmatrix} 3\lambda \\ 5\lambda \end{bmatrix}$$

$\lambda = \frac{13}{3}, \lambda = 2$ not unique

i.e in option (c), 2nd is not an E.vector

Let us take (c)

$$AX = \lambda X$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 7 \\ 0 \end{bmatrix} = \lambda\begin{bmatrix} 7 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 7\lambda \\ 0 \end{bmatrix}$$

$\lambda = 1, \quad 0 = 0$

✓

$$AX = \lambda X$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1/2 \end{bmatrix} = \lambda\begin{bmatrix} 1 \\ 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \lambda \\ \lambda/2 \end{bmatrix}$$

$\lambda = 2, \quad 1 = \frac{\lambda}{2}$

$\lambda = 2, \quad \lambda = 2$

✓

i.e (d) ✓

Today (a)

$$X_1 = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} -7 \\ 0 \end{bmatrix}$$

$\therefore X_1 = -\frac{5}{7} X_2$

$7X_1 + 5X_2 = 0$

So X_1 & X_2 are (LD)

Q. If one pair of LIE vectors of $A=\begin{bmatrix}1&2\\0&2\end{bmatrix}$ is $\begin{bmatrix}1\\a\end{bmatrix}$ & $\begin{bmatrix}1\\b\end{bmatrix}$ then $a+b=$?

(Tr. I) Separately finding E. vectors and then comparing

it will take 10 mins

$A=\begin{bmatrix}1&2\\0&2\end{bmatrix}$

$\lambda=1, X=\begin{bmatrix}1\\0\end{bmatrix}\sim\begin{bmatrix}1\\a\end{bmatrix}\Rightarrow a=0$

$\lambda=2, X=\begin{bmatrix}2\\1\end{bmatrix}\sim\begin{bmatrix}1\\b\end{bmatrix}\Rightarrow b=\frac{1}{2}$

$\approx\begin{bmatrix}1\\1/2\end{bmatrix}$

$\therefore$ Ans $a+b=\frac{1}{2}$

(Method II) ∵ A is U.T.M so $\lambda = 1$ & $\lambda = 2$

Let $X = \begin{bmatrix} 1 \\ a \end{bmatrix}$ & $X = \begin{bmatrix} 1 \\ b \end{bmatrix}$

Then By Concept, $AX = \lambda X$ &

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ a \end{bmatrix} = 1\begin{bmatrix} 1 \\ a \end{bmatrix}$$

$$\begin{bmatrix} 1+2a \\ 2a \end{bmatrix} = \begin{bmatrix} 1 \\ a \end{bmatrix}$$

$1+2a = 1 \Rightarrow a = 0$

$2a = a \Rightarrow a = 0$

$AX = \lambda X$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ b \end{bmatrix} = 2\begin{bmatrix} 1 \\ b \end{bmatrix}$$

$$\begin{bmatrix} 1+2b \\ 2b \end{bmatrix} = \begin{bmatrix} 2 \\ 2b \end{bmatrix}$$

$1+2b = 2$, $2b = 2b$ $\Rightarrow b = \frac{1}{2}$

Hence $a+b = \frac{1}{2}$ Ans

SIMILAR MATRICES → Two Matrices A & B are called SIMILAR if it is possible to find a relationship of the type

$$P^{-1}AP = B$$

where P is any Invertible Mat & this transformation from A to B is called Similarity Transformation.

Note:- ① Similarity Relation b/n the Matrices is an equivalence relation. i.e (Reflexive, Symmetric & Transitive)

② SIMILAR MATRICES have Same Trace, Same Det, Same C Eqn, & Same Eigen Values.

DIAGONALISATION →

"Sq. Mat A is called Diagonalizable if it is similar to Diag Matrix"

Consider $A_{3\times3}$ s.t it's → Evalues = $\lambda_1, \lambda_2, \lambda_3$

→ Evectors = X_1, X_2, X_3

then Modal Mat $P = [X_1 X_2 X_3]$ & Let $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

Now $\boxed{P^{-1}AP = \cdots\cdots = D}$ i.e A is similar to Diag Mat

① H. Cond^{n} for Diagonalisation →

Number of L.I Eveectors = order of A

Let us take $A_{3\times3}$ & it has only two L-I Eveectors.

then Modal Mat $P=[X_1\ X_2\ X_3]$ will be singular ($\because X_1, X_2, X_3$ are LD)

$\Rightarrow P^{-1}=DNE$

& whole Concept of Diagonalisation will be meaningless.

② Applications of this Concept → we have $P^{-1}AP=D \Rightarrow \boxed{A=PDP^{-1}}$

$P.(P^{-1}AP)P^{-1}=PDP^{-1}$

$A=PDP^{-1}$

Ans

Q: If for the mat $A_{2\times2}$ we have $A\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}6\\6\end{bmatrix}$ & $A\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}-2\\2\end{bmatrix}$ then $A=$?

Sol:

$$A\begin{bmatrix}1\\1\end{bmatrix}=6\begin{bmatrix}1\\1\end{bmatrix} \quad \& \quad A\begin{bmatrix}-1\\1\end{bmatrix}=2\begin{bmatrix}-1\\1\end{bmatrix}$$

$\lambda_1=6,\ X_1=\begin{bmatrix}1\\1\end{bmatrix}$ $\qquad$ $\lambda_2=2,\ X_2=\begin{bmatrix}-1\\1\end{bmatrix}$

$$P=[X_1\ X_2]=\begin{bmatrix}1&-1\\1&1\end{bmatrix},\quad P^{-1}=\frac{1}{2}\begin{bmatrix}1&1\\-1&1\end{bmatrix},\quad D=\begin{bmatrix}6&0\\0&2\end{bmatrix}$$

By using Concept of Diagonalisation,

$$P^{-1}AP=D \Rightarrow A=PDP^{-1}$$

$$=\begin{bmatrix}1&-1\\1&1\end{bmatrix}\begin{bmatrix}6&0\\0&2\end{bmatrix}\frac{1}{2}\begin{bmatrix}1&1\\-1&1\end{bmatrix}=\begin{bmatrix}4&2\\2&4\end{bmatrix}$$

Q: If $A_{2\times 2}$ s.t $A\begin{bmatrix}1\\ i\end{bmatrix}=\begin{bmatrix}i\\ -1\end{bmatrix}$ & $A\begin{bmatrix}i\\ 1\end{bmatrix}=\begin{bmatrix}1\\ -i\end{bmatrix}$ then $A=$?

$$A\begin{bmatrix}1\\ i\end{bmatrix}=i\begin{bmatrix}1\\ i\end{bmatrix} \quad \& \quad A\begin{bmatrix}i\\ 1\end{bmatrix}=-i\begin{bmatrix}i\\ 1\end{bmatrix}$$

i.e. $\lambda_1=i,\ X_1=\begin{bmatrix}1\\ i\end{bmatrix}$ & $\lambda_2=-i,\ X_2=\begin{bmatrix}i\\ 1\end{bmatrix}$

$$P=[X_1 X_2]=\begin{bmatrix}1 & i\\ i & 1\end{bmatrix},\ P^{-1}=\frac{1}{2}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix},\ D=\begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix}$$

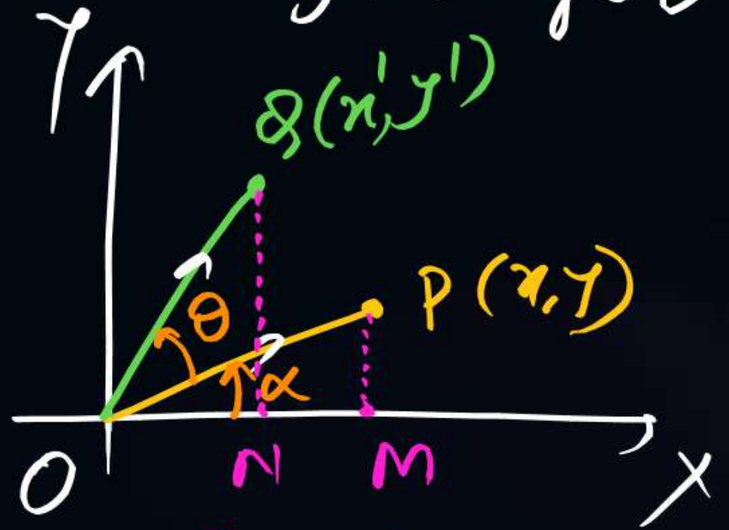
Now $P^{-1}AP=D \Rightarrow A=PDP^{-1}$

$$=\begin{bmatrix}1 & i\\ i & 1\end{bmatrix}\begin{bmatrix}i & 0\\ 0 & -i\end{bmatrix}\frac{1}{2}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix}$$

$$=\begin{bmatrix}i & 1\\ -1 & -i\end{bmatrix}\frac{1}{2}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix}=\frac{1}{2}\begin{bmatrix}0 & 2\\ -2 & 0\end{bmatrix}=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$$

Rotation Mat^{-}

Consider the vector $\begin{bmatrix} x \\ y \end{bmatrix}$ & it is Rotated in an ACW sense by an angle θ and Resultant vector is $\begin{bmatrix} x' \\ y' \end{bmatrix}$



$OM = x$, $PM = y$

$ON = x'$, $QN = y'$

$r = |OP| = |OQ|$

Here $x = r\cos\alpha$, $y = r\sin\alpha$

Again

$x' = r\cos(\theta+\alpha) = r[\cos\theta\cos\alpha - \sin\theta\sin\alpha] = x\cos\theta - y\sin\theta$

$y' = r\sin(\theta+\alpha) = r[\sin\theta\cos\alpha + \cos\theta\sin\alpha] = x\sin\theta + y\cos\theta$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = R \begin{bmatrix} x \\ y \end{bmatrix}$$

where $R = \text{RotMat} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Note If vector is Rotating C-W then RotMat can be obtained by $\theta \to (-\theta)$

Rotation Mat in 3D →

Case I: (when vector is Rotating) →

About X axis

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R_x \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

About Y axis

$$R_y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R_y \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

About Z axis

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = R_z \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Case II: If Coordinate System is Rotating → To find Rotation Mat, Take the Transpose of Rot Matrices obtained in Case I.

Note (1) If vector or Coordinate system is Rotating in C.W sense then To find Rotation Mat, Replace $\theta \longrightarrow -\theta$ in Case I & Case II resp.

(2) $Tr(\text{Rot Mat}) = 1 + 2\cos\theta$

(3) Rotation Matrices are orthogonal Matrices.

If R is a Rot-Mat $\Leftarrow$ $R^T = R^{-1}$, $|R| = 1$

Q: If the vector $\begin{bmatrix}1\\1\end{bmatrix}$ is to be Rotated by an angle 45° in an ACW sense then what will be the Coordinates of new vector ?

Sol: (M-I) $X = \begin{bmatrix}1\\1\end{bmatrix} = \hat{i}+\hat{j} \simeq (1,1)$



$\|x\| = \sqrt{1^2+1^2} = \sqrt{2}$

is Rotated vector $= \begin{bmatrix}0\\\sqrt{2}\end{bmatrix}$

(M-II) $\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$, $\theta = 45^\circ$, $\begin{bmatrix}x'\\y'\end{bmatrix} = ?$

$$\begin{bmatrix}x'\\y'\end{bmatrix} = R\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}$$

$$= \begin{bmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}0\\\frac{2}{\sqrt{2}}\end{bmatrix}$$

$$\begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}0\\\sqrt{2}\end{bmatrix}$$

Q. If the vector $\begin{bmatrix}1\\1\\1\end{bmatrix}$ is to be Rotated about x axis by an angle 30° then find new vector ?

Sol: $\theta = 30°$, $R_x = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos 30° & -\sin 30°\\ 0 & \sin 30° & \cos 30°\end{bmatrix}$

$$\begin{bmatrix}x'\\y'\\z'\end{bmatrix} = R_x \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & \sqrt{3}/2 & -1/2\\ 0 & 1/2 & \sqrt{3}/2\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}1\\ \frac{(\sqrt{3}-1)}{2}\\ \frac{(1+\sqrt{3})}{2}\end{bmatrix}$$ Ans

Very-Very Important Comparison →

① Number of L.I Solutions of Homog system $(AX=0)$ = Nullity of A = No. of Columns − $\rho(A)$

② Number of LI Eigen Vectors for λ = G.M of λ = order − $\rho(A-\lambda I)$

③ Number of L I Column Vectors of A = $\rho(A)$

OR

Number of LI Row Vectors of A = $\rho(A)$

THANK - YOU

Tel: drpuneet_sir_pw